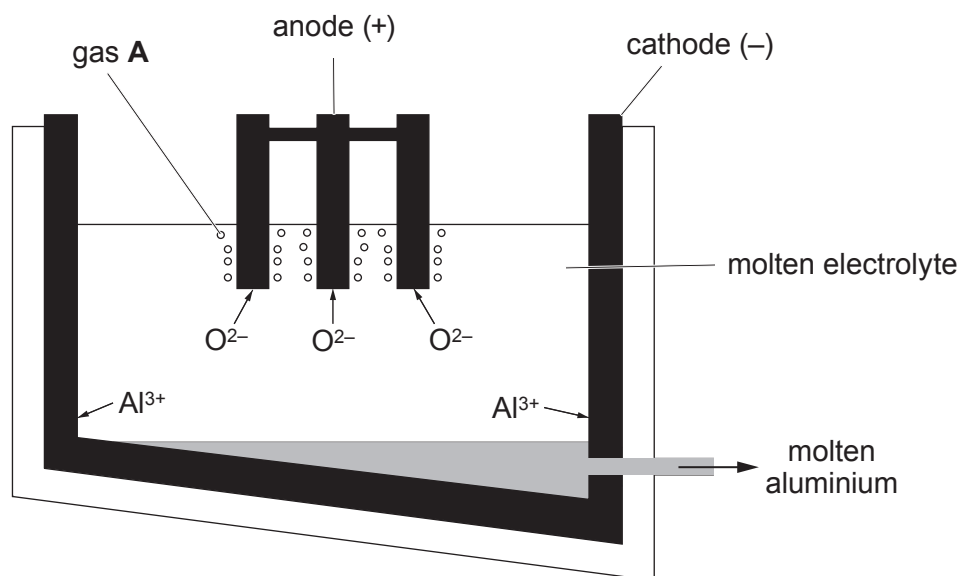


3. (a) The diagram shows a cell used in the extraction of aluminium from aluminium oxide.



positive	oxygen	cryolite	negative
aluminium	hydrogen	steel	graphite

- (i) Choose words from the box to complete each of the following sentences. [4]

The electrolyte is a mixture of molten aluminium oxide and

Aluminium ions, Al^{3+} , are attracted to the electrode with a charge.

Gas **A** is

The electrodes are made of

- (ii) Balance the equation for the overall reaction. [1]



- (b) Calculate the relative formula mass (M_r) of aluminium oxide, Al_2O_3 . [2]

$$A_r(\text{O}) = 16$$

$$A_r(\text{Al}) = 27$$

Relative formula mass =

- (c) The box below contains some properties of aluminium.

low density	non-magnetic	resists corrosion
good thermal conductor	good electrical conductor	

Window frames can be made from different materials.

Choose **two** properties from the box which make aluminium a **better** material than iron for making window frames. [2]

Property 1

Property 2

